Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 40

Fundamental Properties of Trees

Covers Chartrand & Zhang, Section 13.1

Status of this set

Ungraded practice. Solutions released with it.

Part I: How to Practice This Material

Four facts that solve almost everything

1. A tree of order n has size $n-1$. (Thm 13.6)
2. Any *two* of: connected, acyclic, $m = n-1$ — imply the third. (Thm 13.10)
3. A forest of order n with k components has size $n-k$. (Cor 13.8)
4. $n_1 = 2 + n_3 + 2n_4 + 3n_5 + \cdots + (k-2)n_k$. (Thm 13.12)

Two ways to count leaves

Use Theorem 13.12 directly, or set up the degree sum: with order n and size $n-1$, the degrees total $2(n-1)$. Both work; the second is a useful check.

Three mistakes to expect

1. **Assuming $m = n-1$ alone makes a tree.** It needs a second property.
2. **Counting degree-2 vertices in the leaf formula.** They contribute zero.
3. **Forgetting that a forest's size is $n-k$, not $n-1$.**

Part II: Practice Problems

Section A — Sizes and counts

1. Answer each with a number and a one-line reason.

(a) A tree has order 31. What is its size?

(b) A forest has order 20 and 6 components. What is its size?

(c) A connected graph has order 14 and size 22. How many edges must be removed to leave a spanning tree?

2. A tree has three vertices of degree 2, five vertices of degree 4, and no other vertices of degree 2 or more. Find the number of leaves, the order, and the size.

Section B — Existence

3. Decide whether each tree exists. Justify.

(a) A tree of order 100 with every vertex of degree 1 or 3.

(b) A tree of order 100 with every vertex of degree 1 or 5.

Section C — Theorem 13.10

4. For each graph on $V = \{1, 2, 3, 4, 5\}$, say which of the three properties (connected, acyclic, $m = n - 1$) hold, and whether it is a tree.

(a) $E = \{12, 23, 34, 45\}$

(b) $E = \{12, 23, 31, 45\}$

(c) $E = \{12, 13, 14\}$

Section D — Writing

5. In five to eight complete sentences, explain why the proof that every tree of order n has size $n - 1$ needs Theorem 13.5 first. Describe where in the induction the leaf is used.

Section E — Challenge (optional)

6. Prove that if a tree has exactly two leaves, then it is a path.